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Homework 2

Exercise 1

Exercise 2

$\mathbb{L} = \{xx \mid x \in L\}$ for any L?

Yes/no:
No.

Motivation:
xx demands the same $x \in L$ twice, whereas $LL = \{xy \mid x, y \in L\}$, where x and y can be different strings.

If different: find a special case where they are equivalent.
$L = \{a\}$

The only member of LL is aa, which is also satisfies $\{xx \mid x \in L\}$. The same argument holds the other way, therefore the languages are equivalent.

$L_1^R L_2^R = (L_1 L_2)^R$ for all L_1, L_2?

Yes/no:
No.

Motivation:
Consider $L_1 = \{ab\}, L_2 = \{cd\}$. $L_1^R L_2^R = \{badc\}$ but $(L_1 L_2)^R = \{dcba\}$ which are not the same.

If different: find a special case where they are equivalent.
If $L_1 = L_2$ the languages simplify to $L^R L^R$ and $(LL)^R$ which are trivially equal.

$L_1^R L_2^R = (L_2 L_1)^R$ for all L_1, L_2?

Yes/no:
Yes.

Motivation:
Reversing a string st where $s, t \in \Sigma^*$ first switches their order, then reverses the strings themselves, yielding $t^R s^R$. Applying that to our language $L_2^R L_1^R$ thus gives $L_1^{R^R} L_2^{R^R}$, that is equal to $L_1 L_2$.

$L^* = L^* L$ for every L?

Yes/no:
No.

Motivation:
We can choose ε for the L^* part of a string in $L^* L$. Therefore, if $\varepsilon \notin L$, then $\varepsilon \notin L^* L$ so the languages are not equivalent.

If different: find a special case where they are equivalent.

If $\varepsilon \in L$ we can pick ε as the L in L^*L while remaining in the same language (since L^* "subsumes" the L).

Exercise 3

1. State your assumptions on the pumping length $p \geq 0$:

Since we want to prove the statement generally, we cannot make any assumptions on the pumping length p beyond $p \geq 0$.

2. Choose a string $s \in L$ such that $|s| \geq p$:

We choose the string $s = 10^p20^p30^p$. $|s| = 3 + 3p = 3(p+1) > p$.

3. For every decomposition $s = uvwxy$ where $|vx| > 0, |vwx| \leq p$, find a value $i \geq 0$ such that $uv^iwx^iy \notin L_2$:

Because $|vwx| \leq p$ and each number in our string is of length $p+1$, vx cannot cover all three numbers for any decomposition. Therefore, pumping vx will change the value of one or two of these numbers, but not all three, therefore breaking the language constraint. Since we can pump the string out of L, L is not context-free.

Exercise 4

Let G be the following CFG (where S is the start symbol):

$S \to aB \mid aDc$

$B \to bBc \mid c$

$D \to bc \mid c$

(a) Describe the language $L(G)$.

$L(G) = \{ab^nc^nc \mid n > 0\} \cup \{acc\}$

(b) Show that G is ambiguous.

The string $abcc$ can be derived in two ways:

$S \to aB \to abBc \to abcc$

$S \to aDc \to abcc$

(c) Show that G is not an LR(1) grammar.

Let's say we read acc. After reading ac we cannot decide between reducing B or D (with $S \to aB, \{\$\}$ or $S \to aDc, \{\$\}$, respectively).

(d) Modify G into an equivalent grammar G' that is LR(1). Prove that G' is an LR(1) grammar, and explain why G and G' are equivalent.

The main offending production rules are $B \to c$ and $D \to bc$. The ambiguous string $abcc$ is the only string that can use the D production rule, so we can simply remove it without losing any strings of the language:

$S \to aB \mid aDc$

$B \to bBc \mid c$

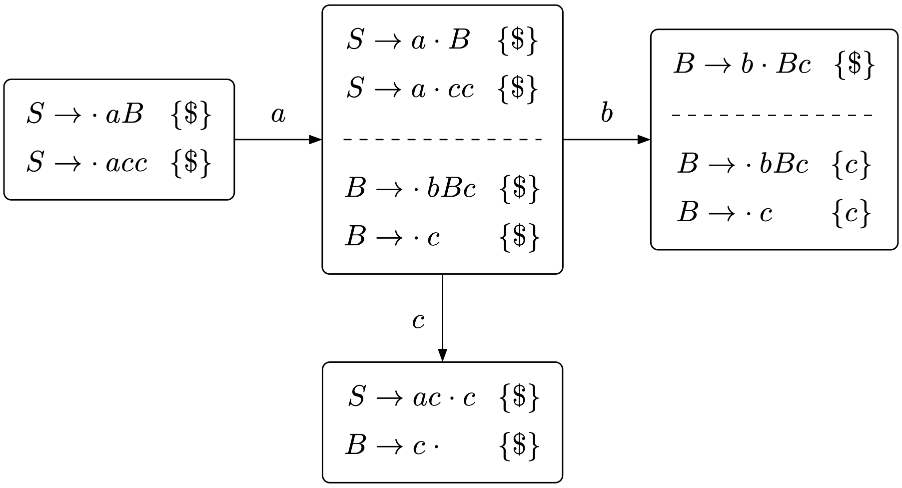
$D \to c$

To simplify we can now remove D as a whole:

$S \rightarrow aB \mid acc$

$B \rightarrow bBc \mid c$

The automaton for this grammar is as follows:



Exercise 5